

Fill in the blanks for completing the structure of AlexNet

| Layer   | Config (F / K / S / P) | Output size                                                                                 | No. of weights                                                                              |
|---------|------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| INPUT   | –                      | 227 × 227 × 3                                                                               | 0                                                                                           |
| CONV1   | F=11, K=96, S=4, P=0   | <input style="border: 1px solid black; width: 100px; height: 20px;" type="text" value="?"/> | <input style="border: 1px solid black; width: 100px; height: 20px;" type="text" value="?"/> |
| POOL1   | Max, F=3, S=2          | <input style="border: 1px solid black; width: 100px; height: 20px;" type="text" value="?"/> | 0                                                                                           |
| CONV2   | F=5, K=256, S=1, P=2   | 27 × 27 × 256                                                                               | <input style="border: 1px solid black; width: 100px; height: 20px;" type="text" value="?"/> |
| POOL2   | Max, F=3, S=2          | <input style="border: 1px solid black; width: 100px; height: 20px;" type="text" value="?"/> | 0                                                                                           |
| CONV3   | F=3, K=384, S=1, P=1   | 13 × 13 × 384                                                                               | 884,736                                                                                     |
| CONV4   | F=3, K=384, S=1, P=1   | 13 × 13 × 384                                                                               | 1,327,104                                                                                   |
| CONV5   | F=3, K=256, S=1, P=1   | 13 × 13 × 256                                                                               | 884,736                                                                                     |
| POOL3   | Max, F=3, S=2          | <input style="border: 1px solid black; width: 100px; height: 20px;" type="text" value="?"/> | 0                                                                                           |
| FLATTEN | –                      | <input style="border: 1px solid black; width: 100px; height: 20px;" type="text" value="?"/> | 0                                                                                           |
| FC1     | 4096 units             | 4096                                                                                        | <input style="border: 1px solid black; width: 100px; height: 20px;" type="text" value="?"/> |
| FC2     | 4096 units             | 4096                                                                                        | 16,777,216                                                                                  |
| FC3     | 1000 units             | 1000                                                                                        | 4,096,000                                                                                   |

**CONV1 – F=11, K=64, S=4, P=0, D\_in=3**

Output W =  $(227 - 11 + 0 + 4) / 4 = 220 / 4 = 55 \rightarrow$  Output size =  $55 \times 55 \times 64$

Weights =  $64 \times 11 \times 11 \times 3 = 64 \times 363 = 23,232$

**POOL1 – F=3, S=2 (no padding in pooling)**

Output W =  $(55 - 3 + 2) / 2 = 54 / 2 = 27 \rightarrow$  Output size =  $27 \times 27 \times 64$  (depth unchanged by pooling)

**CONV2 – F=5, K=256, S=1, P=2, D\_in=64**

Channel of each kernel = 64

Weights =  $256 \times 5 \times 5 \times 64 = 256 \times 1,600 = 409,600$

**POOL2 – F=3, S=2, input =  $27 \times 27 \times 256$**

Output W =  $(27 - 3 + 2) / 2 = 26 / 2 = 13 \rightarrow$  Output size =  $13 \times 13 \times 256$

**POOL3 – F=3, S=2, input =  $13 \times 13 \times 256$**

Output W =  $(13 - 3 + 2) / 2 = 12 / 2 = 6 \rightarrow$  Output size =  $6 \times 6 \times 256$

**FLATTEN – no computation, just reshape**

$6 \times 6 \times 256 = 9,216$  (flattened vector fed to FC layers)

**FC1 – 9,216 inputs → 4,096 outputs**  
**= 9,216 × 4,096 = 37,748,736**

**Weights**